# Linear Algebra I — Week 5

## Bases and Dimension

Name: _____ Date: _____ All work **by hand**: no calculator.

### Key Facts

**Basis.** A subset $S$ of a vector space $V$ is a *basis* if it is linearly independent and spans $V$. Equivalently, every vector of $V$ can be written *uniquely* as a linear combination of the vectors of $S$ (spanning gives existence, independence gives uniqueness). Independence and dimension depend on the field $F$ of scalars ($\mathbb{C}^2$ has dimension 2 over $\mathbb{C}$ but 4 over $\mathbb{R}$).
**Dimension.** $V$ is *finite-dimensional* if it has a finite basis; then every basis has the same number of elements, called $\dim V$. Standard bases: $\{e_1, \ldots, e_n\}$ for $F^n$ ($\dim n$); $\{E_{ij}\}$ for $M_{m\times n}(F)$ ($\dim mn$); $\{1, x, \ldots, x^n\}$ for $P_n(F)$ ($\dim n+1$). $P(\mathbb{R})$, all polynomials, is not finite-dimensional.
**Dimension Bound.** If $V$ is spanned by $n$ vectors, every linearly independent set in $V$ has at most $n$ elements. So in a space with $\dim V = n$: more than $n$ vectors are linearly dependent; fewer than $n$ vectors cannot span.
**Extension Lemma.** If $S$ is linearly independent and $x \notin S$, then $S \cup \{x\}$ is linearly independent iff $x \notin \text{span}(S)$. A linearly independent set can be extended to a basis by adding vectors from a spanning set one at a time, each outside the current span; a finite spanning set can be shrunk to a basis by discarding vectors lying in the span of the others.
**Basis criterion.** If $\dim V = n$ and $S$ has $n$ elements, then $S$ is a basis iff $S$ is linearly independent iff $S$ spans $V$. If $W$ is a subspace of $V$ then $\dim W \leq \dim V$, with equality only if $W = V$.
**Computing.** Translate vectors, polynomials and matrices into coordinate vectors with respect to a standard basis, then row reduce. The pivot columns of the *original* matrix are a basis of the span of the columns; each non-pivot column of the RREF gives the coefficients of that column in terms of the pivot columns. A subspace defined by linear conditions on the coordinates is the solution set of a homogeneous system: its dimension is the number of free variables, and the vectors multiplying the free parameters form a basis.

## Worked Examples

### Example 1 · Standard bases and dimensions

Prove that $\{1, x, x^2\}$ is a basis of $P_2(F)$ and that $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ is a basis of $M_2(F)$, for any field $F$. State the dimensions.

**Solution.**

1. $P_2(F)$ consists of the polynomials $a+bx+cx^2$ with $a, b, c \in F$. *Spanning:* each such polynomial is $a \cdot 1 + b \cdot x + c \cdot x^2$, a combination of $1, x, x^2$.
2. *Independence:* if $a\cdot 1+b\,x+c\,x^2$ is the zero polynomial, all its coefficients are 0, so $a = b = c = 0$. Hence $\{1, x, x^2\}$ is a basis and $\dim P_2(F) = 3$. In general $\dim P_n(F) = n+1$.
3. $M_2(F)$: $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = aE_{11} + bE_{12} + cE_{21} + dE_{22}$, so the four matrices span. If $aE_{11} + bE_{12} + cE_{21} + dE_{22} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is the zero matrix, then $a = b = c = d = 0$, so they are independent.
4. So $\dim M_2(F) = 4$. In general $\dim M_{m\times n}(F) = mn$, $\dim F^n = n$. None of these dimensions depends on which field $F$ is used (as long as the scalars are $F$).

**Answer:** $\dim P_2(F) = 3$ with basis $\{1, x, x^2\}$; $\dim M_2(F) = 4$ with basis $\{E_{ij}\}$.

### Example 2 · Is a set a basis of $\mathbb{R}^3$?

Decide whether $\{(1,2,0),(0,1,1),(1,0,-2)\}$ is a basis of $\mathbb{R}^3$. If not, describe what it spans.

**Solution.**

1. Put the vectors in the columns: $A = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \end{pmatrix}$. Reduce.
2. Apply $R_2 \to R_2 - 2R_1$:
$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{pmatrix}$$
3. Apply $R_3 \to R_3 - R_2$:
$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$
4. Pivots are in columns 1 and 2 only; column 3 has no pivot, so the vectors are linearly dependent. Column 3 of the RREF is $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$, so $v_3 = 1\cdot v_1 - 2\cdot v_2$. Check: $(1,2,0) - 2(0,1,1) = (1,0,-2)$ ✓.
5. Three vectors that are dependent cannot be a basis. Since $v_1, v_2$ are independent (pivot columns) and span the same space, the set spans a 2-dimensional subspace (a plane) of $\mathbb{R}^3$, with basis $\{(1,2,0),(0,1,1)\}$.

**Answer:** Not a basis: $v_3 = v_1 - 2v_2$. The span is a plane with basis $\{(1,2,0),(0,1,1)\}$.

### Example 3 · Basis of a constraint subspace

Find a basis and the dimension of $W = \{p \in P_3(\mathbb{R}) :\ p(1) = 0,\ p(-1) = 0\}$.

**Solution.**

1. Write $p(x) = a + bx + cx^2 + dx^3$, coordinates $(a,b,c,d)$. Then $p(1) = a + b + c + d$ and $p(-1) = a - b + c - d$. The conditions give the homogeneous system $\begin{cases} a+b+c+d = 0 \\ a-b+c-d = 0 \end{cases}$.
2. Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:
$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & -1 & 1 & -1 & 0 \end{array}\right)$$
3. Apply $R_2 \to R_2 - R_1$:
$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & -2 & 0 & -2 & 0 \end{array}\right)$$
4. Apply $R_2 \to \dfrac{1}{-2} R_2$:
$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array}\right)$$
5. Apply $R_1 \to R_1 - R_2$:
$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array}\right)$$

6. Pivot columns are the columns of the basic variables: $a$, $b$. The remaining variable(s) $c$, $d$ are *free*: call them $c = s$, $d = t$.
7. Read each nonzero row as an equation for its basic variable:

   $a = -s$

   $b = -t$

   $c = s$

   $d = t$
8. Write this as a vector: $\mathbf{x} = s\begin{pmatrix}-1\\0\\1\\0\end{pmatrix} + t\begin{pmatrix}0\\-1\\0\\1\end{pmatrix}$ (the parameters $s$, $t$ may be any numbers).
9. Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix}-1\\-1\\1\\1\end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0, 0)$ ✓.
10. Translate each coordinate vector back into the original objects. Setting one parameter to 1 and the others to 0 in the vector form gives: $\begin{pmatrix}-1\\0\\1\\0\end{pmatrix} \leftrightarrow -1 + x^2$; $\begin{pmatrix}0\\-1\\0\\1\end{pmatrix} \leftrightarrow -x + x^3$.
11. These 2 vectors *span* $W$: the vector form says every element of $W$ equals $s\,\mathbf{b}_1 + t\,\mathbf{b}_2$ with the parameters free. They are *linearly independent*: in this combination the entry in the position of the free variables $c$, $d$ equals the parameters $s$, $t$ (each basis vector has 1 in its own free position and 0 in the other free positions), so a combination equal to $\mathbf{0}$ forces every parameter to be 0.
12. Hence the objects above form a basis of $W$, and $\dim W$ equals the number of free variables, namely 2.
13. Check: $q = -1 + x^2$ has $q(1) = 0$ and $q(-1) = 0$ ✓; $r = -x + x^3$ has $r(1) = 0$ and $r(-1) = 0$ ✓.

**Answer:** Basis: $\{-1 + x^2,\ -x + x^3\}$; $\dim W = 2$.

## Example 4 · Extracting a basis from a spanning set

The set $S = \{(1,1,0), (0,1,1), (1,2,1), (1,0,-1)\}$ spans a subspace $U$ of $\mathbb{R}^3$. Find a basis of $U$ chosen from $S$, and write the other vectors through it.

**Solution.**

1. The columns of $A = \begin{pmatrix}1&0&1&1\\1&1&2&0\\0&1&1&-1\end{pmatrix}$ are the vectors of $S$. Reduce.
2. Apply $R_2 \to R_2 - R_1$:

   $\begin{pmatrix}1&0&1&1\\0&1&1&-1\\0&1&1&-1\end{pmatrix}$
3. Apply $R_3 \to R_3 - R_2$:

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

4. Pivot columns 1 and 2. Rule: the corresponding columns of the *original* matrix form a basis of the span. So $\{(1,1,0),(0,1,1)\}$ is a basis of $U$ and $\dim U = 2$.
5. Column 3 of the RREF is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$: $v_3 = v_1 + v_2$; column 4 is $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$: $v_4 = v_1 - v_2$. Check: $(1,1,0)+(0,1,1) = (1,2,1)$ ✓and $(1,1,0)-(0,1,1) = (1,0,-1)$ ✓.

**Answer:** Basis $\{(1,1,0),(0,1,1)\}$, $\dim U = 2$; $v_3 = v_1 + v_2$, $v_4 = v_1 - v_2$.

## Example 5 · Extending an independent set to a basis

Extend the linearly independent set $\{(1,1,1),(1,2,3)\}$ to a basis of $\mathbb{R}^3$ using standard basis vectors.

**Solution.**

1. Put $v_1, v_2$ followed by $e_1, e_2, e_3$ in the columns and reduce; keep the pivot columns (Extension Lemma: each kept vector is outside the span of those before it). $B = \left(\begin{array}{cc|ccc} 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 0 & 1 \end{array}\right)$.
2. Apply $R_2 \to R_2 - R_1,\ R_3 \to R_3 - R_1$:

$$\left(\begin{array}{cc|ccc} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 2 & -1 & 0 & 1 \end{array}\right)$$

3. Apply $R_1 \to R_1 - R_2,\ R_3 \to R_3 - 2R_2$:

$$\left(\begin{array}{cc|ccc} 1 & 0 & 2 & -1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \end{array}\right)$$

4. Apply $R_1 \to R_1 - 2R_3,\ R_2 \to R_2 + R_3$:

$$\left(\begin{array}{cc|ccc} 1 & 0 & 0 & 3 & -2 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & -2 & 1 \end{array}\right)$$

5. The pivot columns are $1, 2, 3$. Column 3 is $e_1$, so $\{(1,1,1),(1,2,3),(1,0,0)\}$ is a basis (three independent vectors in $\mathbb{R}^3$). Columns 4 and 5 ($e_2, e_3$) have no pivot: each is already in the span of the previous vectors.
6. Direct check that $e_1 \notin \operatorname{span}\{v_1, v_2\}$: $a(1,1,1) + b(1,2,3) = (1,0,0)$ needs $a + 2b = 0$ and $a + 3b = 0$, so $b = 0$, $a = 0$, but then the first coordinate is $0 \neq 1$. Not in the span ✓.

**Answer:** $\{(1,1,1),(1,2,3),(1,0,0)\}$ is a basis of $\mathbb{R}^3$.

## Example 6 · Coordinates in $\mathbb{C}^2$

Show that $\beta = \{(1,i),(1,-i)\}$ is a basis of $\mathbb{C}^2$ over $\mathbb{C}$ and find $[w]_\beta$ for $w = (3, 1+i)$.

**Solution.**

1. We need $a(1,i) + b(1,-i) = (3, 1+i)$, i.e. $a + b = 3$ and $ia - ib = 1 + i$. The matrix has the basis vectors as columns.
2. Write the system as an augmented matrix, one row per equation, with a bar before the

right-hand side:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ i & -i & 1+i \end{array}\right)$$

3. Apply $R_2 \to R_2 - iR_1$:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -2i & 1-2i \end{array}\right)$$

4. Apply $R_2 \to \dfrac{1}{-2i}R_2$:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1+\frac{1}{2}i \end{array}\right)$$

5. Apply $R_1 \to R_1 - R_2$:

$$\left(\begin{array}{cc|c} 1 & 0 & 2-\frac{1}{2}i \\ 0 & 1 & 1+\frac{1}{2}i \end{array}\right)$$

6. Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

7. Read each nonzero row as an equation for its basic variable:

$$a = 2 - \frac{1}{2}i$$
$$b = 1 + \frac{1}{2}i$$

8. Check by substituting $(a,b) = (2-\frac{1}{2}i, 1+\frac{1}{2}i)$ into the original equations: the left sides give $(3, 1+i)$, the right-hand sides ✓.

9. The coefficient part reduced to $I_2$ (a pivot in every row and column), so for every $w$ there is exactly one solution: $\beta$ spans and is independent, hence is a basis, and coordinates are unique.

10. So $[w]_\beta = (2-\frac{1}{2}i,\ 1+\frac{1}{2}i)^T$. Check: $a+b=3$ ✓and $i(a-b) = i(1-i) = i+1$ ✓.

**Answer:** $\beta$ is a basis and $[w]_\beta = \begin{pmatrix} 2-\frac{1}{2}i \\ 1+\frac{1}{2}i \end{pmatrix}$.

## Example 7 · Counting arguments

Decide using dimensions only: (a) can $\{1, x\}$ span $P_2(\mathbb{R})$? (b) are three vectors in $\mathbb{C}^2$ (over $\mathbb{C}$) necessarily dependent? (c) can a set of 6 vectors be a basis of $M_{2\times 3}(\mathbb{R})$? (d) can 4 independent vectors lie in a subspace of $\mathbb{R}^5$ of dimension 3?

**Solution.**

1. (a) $\dim P_2(\mathbb{R}) = 3$ and a spanning set has at least 3 vectors. $\{1, x\}$ has 2, so it cannot span.
2. (b) $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$, so $\mathbb{C}^2$ is spanned by 2 vectors and every independent set has at most 2 elements. So any 3 vectors are dependent: yes.
3. (c) $\dim M_{2\times 3}(\mathbb{R}) = 6$, so a basis has exactly 6 elements. A set of 6 vectors *can* be a basis (it is a basis iff it is independent, or iff it spans); the count alone does not decide it.
4. (d) A subspace $U$ with $\dim U = 3$ is spanned by 3 vectors, so every independent set in $U$ has at most 3 elements. 4 independent vectors cannot lie in $U$: no.

**Answer:** (a) No. (b) Yes. (c) Possibly (count is consistent; check independence). (d) No.

### Example 8 · Proof: the Extension Lemma

Let $S$ be linearly independent in $V$ and $x \in V$ with $x \notin S$. Prove: $S \cup \{x\}$ is linearly independent if and only if $x \notin \operatorname{span}(S)$.

**Solution.**

1. ($\Rightarrow$) We prove the contrapositive. Suppose $x \in \operatorname{span}(S)$: $x = c_1 s_1 + \cdots + c_m s_m$ with $s_i \in S$. Then $1 \cdot x - c_1 s_1 - \cdots - c_m s_m = \mathbf{0}$ is a relation among distinct elements of $S \cup \{x\}$ with coefficient $1 \neq 0$ on $x$. So $S \cup \{x\}$ is dependent.
2. ($\Leftarrow$) Suppose $x \notin \operatorname{span}(S)$ and $\alpha x + c_1 s_1 + \cdots + c_m s_m = \mathbf{0}$ with $s_i \in S$ distinct. If $\alpha \neq 0$, divide by $\alpha$ (possible in a field): $x = -\alpha^{-1}(c_1 s_1 + \cdots + c_m s_m) \in \operatorname{span}(S)$, contradicting $x \notin \operatorname{span}(S)$.
3. So $\alpha = 0$, leaving $c_1 s_1 + \cdots + c_m s_m = \mathbf{0}$. Since $S$ is independent, every $c_i = 0$. All coefficients are 0, so $S \cup \{x\}$ is independent.

**Answer:** A relation involving $x$ with $\alpha \neq 0$ would put $x$ in $\operatorname{span} S$; if $x \notin \operatorname{span} S$ then $\alpha = 0$ and independence of $S$ kills the rest.

## Practice Problems

### Part A — Standard Bases, Counting and Dimension

**1.** [Foundational]

State the standard basis and the dimension of each space: (a) $\mathbb{C}^3$ over $\mathbb{C}$; (b) $M_{2\times 3}(\mathbb{R})$; (c) $P_3(\mathbb{R})$. Then write $A = \begin{pmatrix} 2 & -1 & 0 \\ 5 & 3 & 4 \end{pmatrix}$ and $p(x) = 4 - x + 2x^3$ as combinations of the standard bases of (b) and (c).

**Step 1.** (a) The standard basis of $\mathbb{C}^3$ is $\{e_1, e_2, e_3\}$ with $e_1 = (1,0,0)$, $e_2 = (0,1,0)$, $e_3 = (0,0,1)$. It has 3 elements, so $\dim \mathbb{C}^3 = 3$ over $\mathbb{C}$.

**Step 2.** (b) Let $E_{ij}$ be the $2 \times 3$ matrix with 1 in row $i$, column $j$ and 0 elsewhere. The standard basis is $\{E_{11}, E_{12}, E_{13}, E_{21}, E_{22}, E_{23}\}$: $2 \cdot 3 = 6$ matrices. So $\dim M_{2\times 3}(\mathbb{R}) = 6$.

**Step 3.** (c) The standard basis of $P_3(\mathbb{R})$ (polynomials of degree at most 3) is $\{1, x, x^2, x^3\}$. It has 4 elements, so $\dim P_3(\mathbb{R}) = 4$ (always degree bound plus 1).

**Step 4.** Write $A$ entry by entry: $A = 2E_{11} + (-1)E_{12} + 0\,E_{13} + 5E_{21} + 3E_{22} + 4E_{23}$. The coefficients are the entries of $A$ read row by row.

**Step 5.** Write $p$ by its coefficients: $p = 4 \cdot 1 + (-1)\,x + 0\,x^2 + 2\,x^3$.

**Answer:** (a) $\{e_1, e_2, e_3\}$, dim 3. (b) $\{E_{ij}\}$ (six matrices), dim 6; $A = 2E_{11} - E_{12} + 5E_{21} + 3E_{22} + 4E_{23}$. (c) $\{1, x, x^2, x^3\}$, dim 4; $p = 4 \cdot 1 - x + 2x^3$.

**2.** [Foundational]

For each statement decide whether it is true, using only a dimension count (state the dimension used and the theorem): (a) five matrices in $M_2(\mathbb{R})$ must be linearly dependent; (b) $\{(1,0,0), (0,1,0)\}$ spans $\mathbb{R}^3$; (c) any four polynomials in $P_2(\mathbb{R})$ are linearly dependent; (d) five matrices in $M_{3\times 2}(\mathbb{R})$ can span $M_{3\times 2}(\mathbb{R})$; (e) ten matrices in $M_3(\mathbb{R})$ can form a basis of $M_3(\mathbb{R})$; (f) three polynomials in $P_3(\mathbb{R})$ can span $P_3(\mathbb{R})$.

**Step 1.** Tools. If $\dim V = n$ then $V$ is spanned by $n$ vectors, so (Dimension Bound) every linearly independent set in $V$ has at most $n$ elements; equivalently, more than $n$ vectors are linearly dependent. Also a spanning set needs at least $n$ vectors (a spanning set can be shrunk to a basis, which has exactly $n$). A basis has exactly $n$ elements.

**Step 2.** (a) $\dim M_2(\mathbb{R}) = 4$. Five vectors is more than 4, so they are linearly dependent. **True.**

**Step 3.** (b) $\dim \mathbb{R}^3 = 3$ but the set has only $2 < 3$ vectors, and a spanning set needs at least 3. **False.** (Concretely every combination has third coordinate 0.)

**Step 4.** (c) $\dim P_2(\mathbb{R}) = 3$, so $P_2(\mathbb{R})$ is spanned by 3 vectors and every independent set has at most 3 elements. Four polynomials in $P_2(\mathbb{R})$ therefore cannot be independent. **True.**

**Step 5.** (d) $\dim M_{3\times 2}(\mathbb{R}) = 3 \cdot 2 = 6$. Five vectors is fewer than 6, so they cannot span. **False.**

**Step 6.** (e) $\dim M_3(\mathbb{R}) = 9$. A basis has exactly 9 elements, not 10. **False.**

**Step 7.** (f) $\dim P_3(\mathbb{R}) = 4$. Three vectors is fewer than 4, so they cannot span. **False.**

**Answer:** (a) true; (b) false; (c) true; (d) false; (e) false; (f) false.

**3.** [Foundational]

Is $\{(1,1,0),\ (1,2,1),\ (0,1,3)\}$ a basis of $\mathbb{R}^3$? Row reduce the matrix whose columns are these vectors.

**Step 1.** Put the vectors in the columns: $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}$. The set is linearly independent exactly when $A\mathbf{a} = \mathbf{0}$ has only $\mathbf{a} = \mathbf{0}$, i.e. when every column has a pivot.

**Step 2.** Apply $R_2 \to R_2 - R_1$:

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 3 \end{pmatrix}$$

**Step 3.** Apply $R_1 \to R_1 - R_2,\ R_3 \to R_3 - R_2$:

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

**Step 4.** Apply $R_3 \to \frac{1}{2}R_3$:

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

**Step 5.** Apply $R_1 \to R_1 + R_3,\ R_2 \to R_2 - R_3$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

**Step 6.** The RREF is the identity matrix, so every column has a pivot: the three vectors are linearly independent.

**Step 7.** $\dim \mathbb{R}^3 = 3$ and we have 3 linearly independent vectors, so they form a basis (an independent set with $\dim V$ elements is a basis). Equivalently, a pivot in every *row* means $A\mathbf{a} = \mathbf{b}$ is solvable for every $\mathbf{b}$, so they span.

**Answer:** Yes, it is a basis of $\mathbb{R}^3$ (RREF $= I_3$).

**4.** [Intermediate]

(Dimension depends on the scalars.) (a) Find a basis and the dimension of $\mathbb{C}^2$ regarded as a vector space over $\mathbb{R}$, and prove your set is a basis. (b) What is $\dim \mathbb{C}^2$ over $\mathbb{C}$? (c) Why is $\{(1,0),(i,0)\}$ linearly independent over $\mathbb{R}$ but linearly dependent over $\mathbb{C}$?

**Step 1.** (a) Candidate: $\mathcal{B} = \{(1,0),(i,0),(0,1),(0,i)\}$. Over $\mathbb{R}$ the scalars are real numbers only.

**Step 2.** *Spanning.* Any $(z,w) \in \mathbb{C}^2$ has $z = a + bi$ and $w = c + di$ with $a,b,c,d \in \mathbb{R}$. Then $(z,w) = a(1,0) + b(i,0) + c(0,1) + d(0,i)$, a combination with real coefficients.

**Step 3.** *Independence.* Suppose $a(1,0) + b(i,0) + c(0,1) + d(0,i) = (0,0)$ with $a,b,c,d \in \mathbb{R}$. First coordinate: $a + bi = 0$; a complex number $a + bi$ with $a,b$ real is 0 only when $a = 0$ and $b = 0$. Second coordinate: $c + di = 0$, so $c = d = 0$. Only the trivial combination gives $\mathbf{0}$.

**Step 4.** So $\mathcal{B}$ is a basis with 4 elements: $\dim_{\mathbb{R}} \mathbb{C}^2 = 4$.

**Step 5.** (b) Over $\mathbb{C}$, $(z,w) = z(1,0) + w(0,1)$ with *complex* coefficients $z,w$, and $\{(1,0),(0,1)\}$ is independent (if $z(1,0) + w(0,1) = \mathbf{0}$ then $z = w = 0$). So $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$.

**Step 6.** (c) Over $\mathbb{R}$: $a(1,0) + b(i,0) = (a + bi, 0) = \mathbf{0}$ forces $a = b = 0$ (part (a)), so independent. Over $\mathbb{C}$: $i \cdot (1,0) - 1 \cdot (i,0) = (i,0) - (i,0) = \mathbf{0}$ is a combination with coefficients $i$ and $-1$, not both

zero, so dependent. The allowed scalars decide independence, so the dimension depends on the field.

**Answer:** (a) Basis $\{(1,0),(i,0),(0,1),(0,i)\}$, $\dim_{\mathbb{R}} = 4$. (b) $\dim_{\mathbb{C}} = 2$. (c) $i(1,0) - (i,0) = \mathbf{0}$ is a nontrivial *complex* relation, but no nontrivial real one exists.

## Part B — Bases of Subspaces Given by Constraints

**5.** [Foundational]

Let $W = \{(x,y,z) \in \mathbb{R}^3 : \ x = 2y\}$. Show that $W$ is a subspace by exhibiting a basis, and state $\dim W$.

**Step 1.** A vector $(x,y,z)$ lies in $W$ exactly when it satisfies the one linear equation $x - 2y = 0$, i.e. $x - 2y + 0z = 0$. So $W$ is the solution set of the homogeneous system $\begin{cases} x - 2y = 0 \end{cases}$ with coefficient matrix $\begin{pmatrix} 1 & -2 & 0 \end{pmatrix}$.

**Step 2.** Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \end{array}\right)$$

**Step 3.** The matrix is already in reduced row echelon form.

**Step 4.** Pivot columns are the columns of the basic variables: $x$. The remaining variable(s) $y$, $z$ are *free*: call them $y = s$, $z = t$.

**Step 5.** Read each nonzero row as an equation for its basic variable:

$$x = 2s$$
$$y = s$$
$$z = t$$

**Step 6.** Write this as a vector: $\mathbf{x} = s\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ (the parameters $s$, $t$ may be any numbers).

**Step 7.** Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0)$ ✓.

**Step 8.** Translate each coordinate vector back into the original objects. Setting one parameter to 1 and the others to 0 in the vector form gives: $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow (2,1,0)$; $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow (0,0,1)$.

**Step 9.** These 2 vectors *span* $W$: the vector form says every element of $W$ equals $s\,\mathbf{b}_1 + t\,\mathbf{b}_2$ with the parameters free. They are *linearly independent*: in this combination the entry in the position of the free variables $y$, $z$ equals the parameters $s$, $t$ (each basis vector has 1 in its own free position and 0 in the other free positions), so a combination equal to $\mathbf{0}$ forces every parameter to be 0.

**Step 10.** Hence the objects above form a basis of $W$, and $\dim W$ equals the number of free variables, namely 2.

**Answer:** Basis: $\{(2,1,0),\ (0,0,1)\}$; $\dim W = 2$.

**6.** [Foundational]

Find a basis and the dimension of $W = \{A \in M_2(\mathbb{R}) : \ \operatorname{tr} A = 0\}$.

**Step 1.** A matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has coordinate vector $(a, b, c, d)$ with respect to $E_{11}, E_{12}, E_{21}, E_{22}$. The condition $\operatorname{tr} A = 0$ reads $a + d = 0$: the homogeneous system $\{a + d = 0$ with coefficient matrix $\begin{pmatrix} 1 & 0 & 0 & 1 \end{pmatrix}$.

**Step 2.** Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \end{array}\right)$$

**Step 3.** The matrix is already in reduced row echelon form.

**Step 4.** Pivot columns are the columns of the basic variables: $a$. The remaining variable(s) $b$, $c$, $d$ are *free*: call them $b = s$, $c = t$, $d = u$.

**Step 5.** Read each nonzero row as an equation for its basic variable:

$$a = -u$$
$$b = s$$
$$c = t$$
$$d = u$$

**Step 6.** Write this as a vector: $\mathbf{x} = s \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + u \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$ (the parameters $s$, $t$, $u$ may be any numbers).

**Step 7.** Check with $s = 1$, $t = 1$, $u = 1$: $\mathbf{x} = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides (0) ✓.

**Step 8.** Translate each coordinate vector back into the original objects. Setting one parameter to 1 and the others to 0 in the vector form gives: $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$; $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$; $\begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

**Step 9.** These 3 vectors *span* $W$: the vector form says every element of $W$ equals $s\,\mathbf{b}_1 + t\,\mathbf{b}_2 + u\,\mathbf{b}_3$ with the parameters free. They are *linearly independent*: in this combination the entry in the position of the free variables $b$, $c$, $d$ equals the parameters $s$, $t$, $u$ (each basis vector has 1 in its own free position and 0 in the other free positions), so a combination equal to $\mathbf{0}$ forces every parameter to be 0.

**Step 10.** Hence the objects above form a basis of $W$, and $\dim W$ equals the number of free variables, namely 3.

**Answer:** Basis: $\left\{ \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right\}$; $\dim W = 3$.

**7.** [Intermediate]

Find a basis and the dimension of $W = \{f \in P_3(\mathbb{R}) : \ f(0) = f(1) = 0\}$.

**Step 1.** Write $f(x) = a + bx + cx^2 + dx^3$; its coordinate vector is $(a, b, c, d)$ with respect to $1, x, x^2, x^3$. Then $f(0) = a$ and $f(1) = a + b + c + d$, so the conditions are $a = 0$ and $a + b + c + d = 0$: the homogeneous system $\begin{cases} a = 0 \\ a + b + c + d = 0 \end{cases}$.

**Step 2.** Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{array}\right)$$

**Step 3.** Apply $R_2 \to R_2 - R_1$:

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{array}\right)$$

**Step 4.** Pivot columns are the columns of the basic variables: $a$, $b$. The remaining variable(s) $c$, $d$ are *free*: call them $c = s$, $d = t$.

**Step 5.** Read each nonzero row as an equation for its basic variable:

$$a = 0$$
$$b = -s - t$$
$$c = s$$
$$d = t$$

**Step 6.** Write this as a vector: $\mathbf{x} = s \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$ (the parameters $s$, $t$ may be any numbers).

**Step 7.** Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix} 0 \\ -2 \\ 1 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0, 0)$ ✓.

**Step 8.** Translate each coordinate vector back into the original objects. Setting one parameter to 1 and the others to 0 in the vector form gives: $\begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow -x + x^2$; $\begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow -x + x^3$.

**Step 9.** These 2 vectors *span* $W$: the vector form says every element of $W$ equals $s\,\mathbf{b}_1 + t\,\mathbf{b}_2$ with the parameters free. They are *linearly independent*: in this combination the entry in the position of the free variables $c$, $d$ equals the parameters $s$, $t$ (each basis vector has 1 in its own free position and 0 in the other free positions), so a combination equal to $\mathbf{0}$ forces every parameter to be 0.

**Step 10.** Hence the objects above form a basis of $W$, and $\dim W$ equals the number of free variables, namely 2.

**Answer:** Basis: $\{-x + x^2,\ -x + x^3\}$; $\dim W = 2$.

**8.** [Intermediate]

Find a basis and the dimension of $W = \{p \in P_3(\mathbb{R}) : \ p(0) = p(1),\ p(2) = 0\}$.

**Step 1.** Write $p(x) = a+bx+cx^2+dx^3$. Then $p(0) = a$, $p(1) = a+b+c+d$ and $p(2) = a+2b+4c+8d$. The condition $p(0) = p(1)$ becomes $b + c + d = 0$ (subtract $a$ from both sides), and $p(2) = 0$ becomes $a+2b+4c+8d = 0$. So we solve the homogeneous system $\begin{cases} b + c + d = 0 \\ a + 2b + 4c + 8d = 0 \end{cases}$.

**Step 2.** Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cccc|c} 0 & 1 & 1 & 1 & 0 \\ 1 & 2 & 4 & 8 & 0 \end{array}\right)$$

**Step 3.** Apply $R_1 \leftrightarrow R_2$:

$$\left(\begin{array}{cccc|c} 1 & 2 & 4 & 8 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{array}\right)$$

**Step 4.** Apply $R_1 \to R_1 - 2R_2$:

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 6 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{array}\right)$$

**Step 5.** Pivot columns are the columns of the basic variables: $a$, $b$. The remaining variable(s) $c$, $d$ are *free*: call them $c = s$, $d = t$.

**Step 6.** Read each nonzero row as an equation for its basic variable:

$$a = -2s - 6t$$
$$b = -s - t$$
$$c = s$$
$$d = t$$

**Step 7.** Write this as a vector: $\mathbf{x} = s\begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} -6 \\ -1 \\ 0 \\ 1 \end{pmatrix}$ (the parameters $s$, $t$ may be any numbers).

**Step 8.** Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix} -8 \\ -2 \\ 1 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0, 0)$ ✓.

**Step 9.** Translate each coordinate vector back into the original objects. Setting one parameter to 1 and the others to 0 in the vector form gives: $\begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow -2 - x + x^2$; $\begin{pmatrix} -6 \\ -1 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow -6 - x + x^3$.

**Step 10.** These 2 vectors *span* $W$: the vector form says every element of $W$ equals $s\,\mathbf{b}_1 + t\,\mathbf{b}_2$ with the parameters free. They are *linearly independent*: in this combination the entry in the position of the free variables $c$, $d$ equals the parameters $s$, $t$ (each basis vector has 1 in its own free position and 0 in the other free positions), so a combination equal to $\mathbf{0}$ forces every parameter to be 0.

**Step 11.** Hence the objects above form a basis of $W$, and $\dim W$ equals the number of free variables, namely 2.

**Step 12.** Check the first basis polynomial $q = -2 - x + x^2$: $q(0) = -2$, $q(1) = -2 - 1 + 1 = -2$, so $q(0) = q(1)$; and $q(2) = -2 - 2 + 4 = 0$ ✓. Check the second $r = -6 - x + x^3$: $r(0) = -6$, $r(1) = -6 - 1 + 1 = -6$, and $r(2) = -6 - 2 + 8 = 0$ ✓.

**Answer:** Basis: $\{-2 - x + x^2,\ -6 - x + x^3\}$; $\dim W = 2$.

**9.** [Intermediate]

Find a basis and the dimension of $W = \{(x, y, z, w) \in \mathbb{R}^4 :\ x + y - z = 0,\ y + w = 0\}$.

**Step 1.** A vector $(x, y, z, w)$ lies in $W$ exactly when $x + y - z = 0$ and $y + w = 0$: the homogeneous system $\begin{cases} x + y - z = 0 \\ y + w = 0 \end{cases}$.

**Step 2.** Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array}\right)$$

**Step 3.** Apply $R_1 \to R_1 - R_2$:

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{array}\right)$$

**Step 4.** Pivot columns are the columns of the basic variables: $x$, $y$. The remaining variable(s) $z$, $w$ are *free*: call them $z = s$, $w = t$.

**Step 5.** Read each nonzero row as an equation for its basic variable:

$$x = s + t$$
$$y = -t$$
$$z = s$$
$$w = t$$

**Step 6.** Write this as a vector: $\mathbf{x} = s \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 0 \\ 1 \end{pmatrix}$ (the parameters $s$, $t$ may be any numbers).

**Step 7.** Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix} 2 \\ -1 \\ 1 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0, 0)$ ✓.

**Step 8.** Translate each coordinate vector back into the original objects. Setting one parameter to 1 and the others to 0 in the vector form gives: $\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow (1, 0, 1, 0)$; $\begin{pmatrix} 1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow (1, -1, 0, 1)$.

**Step 9.** These 2 vectors *span* $W$: the vector form says every element of $W$ equals $s\,\mathbf{b}_1 + t\,\mathbf{b}_2$ with the parameters free. They are *linearly independent*: in this combination the entry in the position of the free variables $z$, $w$ equals the parameters $s$, $t$ (each basis vector has 1 in its own free position and 0 in the other free positions), so a combination equal to $\mathbf{0}$ forces every parameter to be 0.

**Step 10.** Hence the objects above form a basis of $W$, and $\dim W$ equals the number of free variables, namely 2.

**Answer:** Basis: $\{(1,0,1,0),\ (1,-1,0,1)\}$; $\dim W = 2$.

**10.** [Intermediate]

Let $W = \{A \in M_2(\mathbb{C}) :\ a_{11} + i\,a_{12} = 0,\ a_{21} = (1+i)\,a_{22}\}$, a subset of $M_2(\mathbb{C})$ over $\mathbb{C}$. Find a basis of $W$ and $\dim W$.

**Step 1.** Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{C})$. The condition $a_{11} + i\,a_{12} = 0$ reads $a + ib = 0$. The condition $a_{21} = (1+i)a_{22}$ reads $c - (1+i)d = 0$, i.e. $c + (-1-i)d = 0$. The coefficient matrix of this homogeneous system over $\mathbb{C}$ is $\begin{pmatrix} 1 & i & 0 & 0 \\ 0 & 0 & 1 & -1-i \end{pmatrix}$, which is already in RREF.

**Step 2.** Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cccc|c} 1 & i & 0 & 0 & 0 \\ 0 & 0 & 1 & -1-i & 0 \end{array}\right)$$

**Step 3.** The matrix is already in reduced row echelon form.

**Step 4.** Pivot columns are the columns of the basic variables: $a$, $c$. The remaining variable(s) $b$, $d$ are *free*: call them $b = s$, $d = t$.

**Step 5.** Read each nonzero row as an equation for its basic variable:

$$a = -is$$
$$b = s$$
$$c = (1+i)t$$
$$d = t$$

**Step 6.** Write this as a vector: $\mathbf{x} = s\begin{pmatrix} -i \\ 1 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} 0 \\ 0 \\ 1+i \\ 1 \end{pmatrix}$ (the parameters $s$, $t$ may be any numbers).

**Step 7.** Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix} -i \\ 1 \\ 1+i \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0, 0)$ ✓.

**Step 8.** Translate each coordinate vector back into the original objects. Setting one parameter to 1 and the others to 0 in the vector form gives: $\begin{pmatrix} -i \\ 1 \\ 0 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} -i & 1 \\ 0 & 0 \end{pmatrix}$; $\begin{pmatrix} 0 \\ 0 \\ 1+i \\ 1 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 & 0 \\ 1+i & 1 \end{pmatrix}$.

**Step 9.** These 2 vectors *span* $W$: the vector form says every element of $W$ equals $s\,\mathbf{b}_1 + t\,\mathbf{b}_2$ with the parameters free. They are *linearly independent*: in this combination the entry in the position of the free variables $b$, $d$ equals the parameters $s$, $t$ (each basis vector has 1 in its own free position and 0 in the other free positions), so a combination equal to $\mathbf{0}$ forces every parameter to be 0.

**Step 10.** Hence the objects above form a basis of $W$, and $\dim W$ equals the number of free variables, namely 2.

**Step 11.** Check the first basis matrix $\begin{pmatrix} -i & 1 \\ 0 & 0 \end{pmatrix}$: $a_{11} + ia_{12} = -i + i \cdot 1 = 0$ ✓and $a_{21} = 0 = (1+i) \cdot 0$

✓. Check the second $\begin{pmatrix} 0 & 0 \\ 1+i & 1 \end{pmatrix}$: $a_{11} + ia_{12} = 0$ ✓and $a_{21} = 1 + i = (1+i) \cdot 1$ ✓.

**Answer:** Basis: $\left\{ \begin{pmatrix} -i & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1+i & 1 \end{pmatrix} \right\}$; $\dim W = 2$.

**11.** [Intermediate]

Find a basis and the dimension of $W = \{A \in M_2(\mathbb{R}) : \ \operatorname{tr} A = 0 \text{ and } A = A^T\}$.

**Step 1.** Write $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The condition $\operatorname{tr} A = 0$ is $a + d = 0$. The condition $A = A^T$ means $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$, i.e. $b = c$, i.e. $b - c = 0$. So we solve the homogeneous system $\begin{cases} a + d = 0 \\ b - c = 0 \end{cases}$.

**Step 2.** Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 \end{array}\right)$$

**Step 3.** The matrix is already in reduced row echelon form.

**Step 4.** Pivot columns are the columns of the basic variables: $a$, $b$. The remaining variable(s) $c$, $d$ are *free*: call them $c = s$, $d = t$.

**Step 5.** Read each nonzero row as an equation for its basic variable:

$$a = -t$$
$$b = s$$
$$c = s$$
$$d = t$$

**Step 6.** Write this as a vector: $\mathbf{x} = s \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$ (the parameters $s$, $t$ may be any numbers).

**Step 7.** Check with $s = 1$, $t = 1$: $\mathbf{x} = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, and substituting into the original equations gives the right-hand sides $(0, 0)$ ✓.

**Step 8.** Translate each coordinate vector back into the original objects. Setting one parameter to 1 and the others to 0 in the vector form gives: $\begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} \leftrightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; $\begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \leftrightarrow \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

**Step 9.** These 2 vectors *span* $W$: the vector form says every element of $W$ equals $s\,\mathbf{b}_1 + t\,\mathbf{b}_2$ with the parameters free. They are *linearly independent*: in this combination the entry in the position of the free variables $c$, $d$ equals the parameters $s$, $t$ (each basis vector has 1 in its own free position and 0 in the other free positions), so a combination equal to $\mathbf{0}$ forces every parameter to be 0.

**Step 10.** Hence the objects above form a basis of $W$, and $\dim W$ equals the number of free variables, namely 2.

**Answer:** Basis: $\left\{\begin{pmatrix}0&1\\1&0\end{pmatrix}, \begin{pmatrix}-1&0\\0&1\end{pmatrix}\right\}$; $\dim W = 2$.

## Part C — Extracting, Extending and Coordinates

**12.** [Foundational]

Find a linearly independent subset of $S = \{(1,2,1),\ (2,4,2),\ (1,1,0)\} \subseteq \mathbb{R}^3$ with the same span as $S$. State $\dim \operatorname{span} S$.

**Step 1.** Put the vectors in the columns of $A = \begin{pmatrix}1&2&1\\2&4&1\\1&2&0\end{pmatrix}$ and row reduce. The pivot columns of the *original* $A$ give an independent set with the same span as the columns.

**Step 2.** Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$:

$$\begin{pmatrix}1&2&1\\0&0&-1\\0&0&-1\end{pmatrix}$$

**Step 3.** Apply $R_2 \to \dfrac{1}{-1}R_2$:

$$\begin{pmatrix}1&2&1\\0&0&1\\0&0&-1\end{pmatrix}$$

**Step 4.** Apply $R_1 \to R_1 - R_2,\ R_3 \to R_3 + R_2$:

$$\begin{pmatrix}1&2&0\\0&0&1\\0&0&0\end{pmatrix}$$

**Step 5.** The pivot columns are 1 and 3. So $\{v_1, v_3\} = \{(1,2,1),(1,1,0)\}$ is linearly independent and spans the same space as $S$.

**Step 6.** Column 2 of the RREF is $\begin{pmatrix}2\\0\\0\end{pmatrix}$, so $v_2 = 2v_1$: $2(1,2,1) = (2,4,2)$ ✓. Hence $v_2$ adds nothing to the span. The span has 2 vectors in a basis: $\dim \operatorname{span} S = 2$.

**Answer:** $\{(1,2,1),(1,1,0)\}$ (columns 1 and 3); $v_2 = 2v_1$; $\dim \operatorname{span} S = 2$.

**13.** [Intermediate]

In $P_2(\mathbb{R})$ let $S = \{1+x,\ 2+2x,\ x+x^2,\ 1+2x+x^2,\ x^2\}$. Find a linearly independent subset of $S$ with the same span as $S$, express each remaining polynomial through it, and decide whether $S$ spans $P_2(\mathbb{R})$.

**Step 1.** Take coordinates with respect to $1, x, x^2$: $1+x \leftrightarrow (1,1,0)$, $2+2x \leftrightarrow (2,2,0)$, $x+x^2 \leftrightarrow (0,1,1)$, $1+2x+x^2 \leftrightarrow (1,2,1)$, $x^2 \leftrightarrow (0,0,1)$. Coordinates turn linear combinations of polynomials into linear combinations of vectors, so we may row reduce the matrix of these columns.

**Step 2.** $A = \begin{pmatrix}1&2&0&1&0\\1&2&1&2&0\\0&0&1&1&1\end{pmatrix}$.

**Step 3.** Apply $R_2 \to R_2 - R_1$:

$$\begin{pmatrix}1&2&0&1&0\\0&0&1&1&0\\0&0&1&1&1\end{pmatrix}$$

**Step 4.** Apply $R_3 \to R_3 - R_2$:

$$\begin{pmatrix}1&2&0&1&0\\0&0&1&1&0\\0&0&0&0&1\end{pmatrix}$$

**Step 5.** The pivot columns are $1, 3, 5$. So a linearly independent subset with the same span is $\{1 + x,\ x + x^2,\ x^2\}$ (columns $1, 3, 5$ of the *original* matrix).

**Step 6.** Non-pivot columns: column 2 of $R$ is $(2, 0, 0)^T$, so $2 + 2x = 2(1 + x)$. Column 4 of $R$ is $(1, 1, 0)^T$, so $1 + 2x + x^2 = 1 \cdot (1 + x) + 1 \cdot (x + x^2)$. Check: $(1 + x) + (x + x^2) = 1 + 2x + x^2$ ✓.

**Step 7.** The span of $S$ has a basis with 3 elements, so $\dim \operatorname{span} S = 3 = \dim P_2(\mathbb{R})$. A subspace of dimension 3 in a 3-dimensional space is the whole space (an independent set of $\dim V$ elements is a basis), so $S$ spans $P_2(\mathbb{R})$.

**Answer:** Independent subset $\{1+x,\ x+x^2,\ x^2\}$; $2+2x = 2(1+x)$ and $1+2x+x^2 = (1+x)+(x+x^2)$. Yes, $S$ spans $P_2(\mathbb{R})$ (and $\{1 + x, x + x^2, x^2\}$ is a basis of it).

**14.** [Intermediate]

Let $S = \{v_1, v_2, v_3\} = \{(1, 2, 3),\ (3, 2, 1),\ (1, 1, 1)\} \subseteq \mathbb{R}^3$. (a) Write the system that tests linear independence and row reduce it. (b) Decide whether $S$ is linearly independent; if not, find an explicit relation. (c) Give a linearly independent subset of $S$ that is a basis of $\operatorname{span} S$, and decide whether $S$ spans $\mathbb{R}^3$.

**Step 1.** (a) $S$ is independent iff $a_1v_1 + a_2v_2 + a_3v_3 = \mathbf{0}$ forces $a_1 = a_2 = a_3 = 0$. In coordinates this is the homogeneous system $\begin{cases} a_1 + 3a_2 + a_3 = 0 \\ 2a_1 + 2a_2 + a_3 = 0 \\ 3a_1 + a_2 + a_3 = 0 \end{cases}$, whose coefficient matrix has columns $v_1, v_2, v_3$: $A = \begin{pmatrix}1&3&1\\2&2&1\\3&1&1\end{pmatrix}$.

**Step 2.** Row reduce $A$.

**Step 3.** Apply $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - 3R_1$:

$$\begin{pmatrix}1&3&1\\0&-4&-1\\0&-8&-2\end{pmatrix}$$

**Step 4.** Apply $R_2 \to \dfrac{1}{-4}R_2$:

$$\begin{pmatrix}1&3&1\\0&1&\frac{1}{4}\\0&-8&-2\end{pmatrix}$$

**Step 5.** Apply $R_1 \to R_1 - 3R_2,\ R_3 \to R_3 + 8R_2$:

$$\begin{pmatrix}1&0&\frac{1}{4}\\0&1&\frac{1}{4}\\0&0&0\end{pmatrix}$$

**Step 6.** (b) The RREF has pivots in columns $1, 2$ only, so $a_3$ is free and the system has nonzero solutions: $S$ is *linearly dependent*.

**Step 7.** Column 3 of the RREF is $\begin{pmatrix} \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \end{pmatrix}$, so $a_1 = -\frac{1}{4}a_3$ and $a_2 = -\frac{1}{4}a_3$. Choose $a_3 = -4$ (to clear fractions): $(a_1, a_2, a_3) = (1, 1, -4)$, giving the relation $v_1 + v_2 - 4v_3 = \mathbf{0}$.

**Step 8.** Check: $(1,2,3) + (3,2,1) - 4(1,1,1) = (4,4,4) - (4,4,4) = (0,0,0)$ ✓.

**Step 9.** (c) The pivot columns of $A$ are 1 and 2, so $\{v_1, v_2\} = \{(1,2,3), (3,2,1)\}$ is linearly independent and spans span $S$ (indeed $v_3 = \frac{1}{4}v_1 + \frac{1}{4}v_2$). So $\dim \operatorname{span} S = 2$.

**Step 10.** Since $\dim \operatorname{span} S = 2 < 3 = \dim \mathbb{R}^3$, $S$ does not span $\mathbb{R}^3$ (a spanning set of $\mathbb{R}^3$ needs at least 3 vectors).

**Answer:** Dependent: $v_1 + v_2 - 4v_3 = \mathbf{0}$. Basis of span $S$: $\{(1,2,3), (3,2,1)\}$, $\dim \operatorname{span} S = 2$, so $S$ does not span $\mathbb{R}^3$.

**15.** [Intermediate]

Extend the linearly independent set $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \right\}$ to a basis of $M_2(\mathbb{R})$ by adding one of $E_{11}, E_{12}, E_{21}, E_{22}$. Justify your choice and say which of the four would *not* work.

**Step 1.** Use coordinates $(a_{11}, a_{12}, a_{21}, a_{22})$: the three matrices are $A_1 = (1,0,0,0)$, $A_2 = (0,1,1,0)$, $A_3 = (0,1,0,1)$, and $E_{11}, E_{12}, E_{21}, E_{22}$ are the standard vectors. $\dim M_2(\mathbb{R}) = 4$, so we need exactly one more vector $x$ with $x \notin \operatorname{span}\{A_1, A_2, A_3\}$ (Extension Lemma: $\{A_1, A_2, A_3, x\}$ is then independent, and 4 independent vectors in a 4-dimensional space form a basis).

**Step 2.** Find $\operatorname{span}\{A_1, A_2, A_3\}$: $\alpha A_1 + \beta A_2 + \gamma A_3$ has coordinates $(\alpha,\ \beta+\gamma,\ \beta,\ \gamma)$. So a matrix $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ lies in the span exactly when $a_{12} = a_{21} + a_{22}$ (the span is $\{(\alpha, \beta+\gamma, \beta, \gamma)\}$; given such coordinates take $\alpha = a_{11}$, $\beta = a_{21}$, $\gamma = a_{22}$).

**Step 3.** Test each standard matrix against $a_{12} = a_{21} + a_{22}$: $E_{11}$: $0 = 0 + 0$ holds, so $E_{11} \in$ span (indeed $E_{11} = A_1$). $E_{12}$: $1 = 0 + 0$ fails, so $E_{12} \notin$ span. $E_{21}$: $0 = 1 + 0$ fails, so not in the span. $E_{22}$: $0 = 0 + 1$ fails, so not in the span.

**Step 4.** Same result by row reduction: put $A_1, A_2, A_3, E_{11}, E_{12}, E_{21}, E_{22}$ as columns of one matrix and reduce; the pivot columns tell which vectors to keep from left to right. $B = \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$.

**Step 5.** Apply $R_3 \to R_3 - R_2$:

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$$

**Step 6.** Apply $R_3 \to \dfrac{1}{-1} R_3$:

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$$

**Step 7.** Apply $R_2 \to R_2 - R_3,\ R_4 \to R_4 - R_3$:

$$\left(\begin{array}{ccc|cccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 1 \end{array}\right)$$

**Step 8.** Apply $R_4 \to \dfrac{1}{-1}R_4$:

$$\left(\begin{array}{ccc|cccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 \end{array}\right)$$

**Step 9.** Apply $R_3 \to R_3 - R_4$:

$$\left(\begin{array}{ccc|cccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 \end{array}\right)$$

**Step 10.** The pivot columns are $1, 2, 3$ and $5$. Column 4 ($E_{11}$) has no pivot because it equals column 1; column 5 ($E_{12}$) is the first standard vector that adds a new direction, so $\{A_1, A_2, A_3, E_{12}\}$ is a basis.

**Step 11.** Conclusion: $E_{12}$, $E_{21}$ and $E_{22}$ each extend the set to a basis; $E_{11}$ does not, since it already lies in the span of the given set.

**Answer:** A basis is $\{A_1, A_2, A_3, E_{12}\}$ (also $E_{21}$ or $E_{22}$ would work); $E_{11}$ fails because $E_{11} = A_1$ is already in the span.

**16.** [Intermediate]

Show that $\beta = \{1 - x,\ 2 + x^2,\ x + 2x^2\}$ is a basis of $P_2(\mathbb{R})$ and find the coordinate vector $[p]_\beta$ of $p(x) = 5 + 2x + 8x^2$.

**Step 1.** With respect to $1, x, x^2$ the three polynomials have coordinate vectors $(1, -1, 0)$, $(2, 0, 1)$, $(0, 1, 2)$, and $p \leftrightarrow (5, 2, 8)$. We need $a(1 - x) + b(2 + x^2) + c(x + 2x^2) = p$, i.e. the system with these vectors as columns and $(5, 2, 8)$ on the right: $a + 2b = 5$, $-a + c = 2$, $b + 2c = 8$ (equating coefficients of $1$, $x$, $x^2$).

**Step 2.** Write the system as an augmented matrix, one row per equation, with a bar before the right-hand side:

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ -1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 8 \end{array}\right)$$

**Step 3.** Apply $R_2 \to R_2 + R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 2 & 1 & 7 \\ 0 & 1 & 2 & 8 \end{array}\right)$$

**Step 4.** Apply $R_2 \to \dfrac{1}{2}R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & \dfrac{1}{2} & \dfrac{7}{2} \\ 0 & 1 & 2 & 8 \end{array}\right)$$

**Step 5.** Apply $R_1 \to R_1 - 2R_2,\ R_3 \to R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & \frac{1}{2} & \frac{7}{2} \\ 0 & 0 & \frac{3}{2} & \frac{9}{2} \end{array}\right)$$

**Step 6.** Apply $R_3 \to \frac{2}{3}R_3$:

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & \frac{1}{2} & \frac{7}{2} \\ 0 & 0 & 1 & 3 \end{array}\right)$$

**Step 7.** Apply $R_1 \to R_1 + R_3,\ R_2 \to R_2 - \frac{1}{2}R_3$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

**Step 8.** Every column of the coefficient part holds a pivot, so there are no free variables and the solution is unique.

**Step 9.** Read each nonzero row as an equation for its basic variable:

$$a = 1$$
$$b = 2$$
$$c = 3$$

**Step 10.** Check by substituting $(a, b, c) = (1, 2, 3)$ into the original equations: the left sides give $(5, 2, 8)$, the right-hand sides ✓.

**Step 11.** The reduction used only the coefficient part, and that part became $I_3$ (a pivot in every column and every row). So for *every* right-hand side the system has exactly one solution: existence of a solution for every polynomial means $\beta$ spans $P_2(\mathbb{R})$, and uniqueness (right-hand side $\mathbf{0}$ gives only $\mathbf{0}$) means $\beta$ is linearly independent. So $\beta$ is a basis and the coordinates are unique.

**Step 12.** Therefore $[p]_\beta = (1, 2, 3)^T$; check: $1(1 - x) + 2(2 + x^2) + 3(x + 2x^2) = (1 + 4) + (-1 + 3)x + (2 + 6)x^2 = 5 + 2x + 8x^2$ ✓.

**Answer:** $\beta$ is a basis, and $[p]_\beta = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

**17.** [Intermediate]

Let $A_1 = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}$, $A_2 = \begin{pmatrix} i & -1 \\ 0 & i \end{pmatrix}$, $A_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ in $M_2(\mathbb{C})$. (a) Over $\mathbb{C}$, decide whether $\{A_1, A_2, A_3\}$ is linearly independent by row reducing, and give a basis of its span. (b) Regard $M_2(\mathbb{C})$ as a vector space over $\mathbb{R}$. Is $\{A_1, A_2, A_3\}$ linearly independent over $\mathbb{R}$? Prove your answer.

**Step 1.** (a) Coordinates $(a_{11}, a_{12}, a_{21}, a_{22})$: $A_1 = (1, i, 0, 1)$, $A_2 = (i, -1, 0, i)$, $A_3 = (0, 1, 1, 0)$. Over $\mathbb{C}$ independence means $\alpha_1 A_1 + \alpha_2 A_2 + \alpha_3 A_3 = \mathbf{0}$ forces $\alpha_1 = \alpha_2 = \alpha_3 = 0$, so put them as columns of $A = \begin{pmatrix} 1 & i & 0 \\ i & -1 & 1 \\ 0 & 0 & 1 \\ 1 & i & 0 \end{pmatrix}$ and reduce (arithmetic over $\mathbb{C}$, using $i^2 = -1$).

**Step 2.** Apply $R_2 \to R_2 - iR_1,\ R_4 \to R_4 - R_1$:

$$\begin{pmatrix} 1 & i & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

**Step 3.** Apply $R_3 \to R_3 - R_2$:

$$\begin{pmatrix} 1 & i & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

**Step 4.** Column 2 has no pivot (pivot columns 1 and 3). Column 2 of the RREF is $\begin{pmatrix} i \\ 0 \\ 0 \\ 0 \end{pmatrix}$, so $A_2 = i\,A_1$.

Check: $i \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} i & i^2 \\ 0 & i \end{pmatrix} = \begin{pmatrix} i & -1 \\ 0 & i \end{pmatrix} = A_2$ ✓.

**Step 5.** So over $\mathbb{C}$ the set is linearly dependent ($iA_1 - A_2 = 0$), and $\{A_1, A_3\}$ (pivot columns $1, 3$) is a basis of the span, of dimension 2.

**Step 6.** (b) Over $\mathbb{R}$ the scalars $x, y, z$ are real. Suppose $xA_1 + yA_2 + zA_3 = 0$ with $x, y, z \in \mathbb{R}$. The $(1,1)$ entry gives $x + iy = 0$. A complex number $x + iy$ with $x, y$ real is 0 only if $x = y = 0$ (compare real and imaginary parts).

**Step 7.** The $(2,1)$ entry gives $z = 0$. So only the trivial combination works: $\{A_1, A_2, A_3\}$ is linearly independent over $\mathbb{R}$.

**Answer:** (a) Dependent over $\mathbb{C}$: $A_2 = iA_1$; basis $\{A_1, A_3\}$ of the span. (b) Independent over $\mathbb{R}$: $x + iy = 0$ with $x, y$ real forces $x = y = 0$, and the $(2,1)$ entry forces $z = 0$.

## Part D — Proofs and Challenges

**18.** [Intermediate]

Let $V$ be a vector space with $\dim V = n$ and let $W$ be a subspace of $V$. Prove: (a) $W$ is finite-dimensional with $\dim W \le n$; (b) if $\dim W = n$ then $W = V$.

**Step 1.** Tools: Dimension Bound (a linearly independent subset of $V$ has at most $n$ elements, since $V$ is spanned by $n$ vectors); Extension Lemma ($S$ independent, $x \notin \operatorname{span} S \Rightarrow S \cup \{x\}$ independent); an independent set of $n$ elements in $V$ is a basis.

**Step 2.** (a) Any linearly independent subset of $W$ is a linearly independent subset of $V$, so it has at most $n$ elements. So the sizes of independent subsets of $W$ are bounded; choose an independent subset $S = \{w_1, \dots, w_m\} \subseteq W$ with $m$ as large as possible, $m \le n$ (the empty set is independent, so such $S$ exists).

**Step 3.** Claim: $S$ spans $W$. If not, there is $x \in W$ with $x \notin \operatorname{span} S$. By the Extension Lemma $S \cup \{x\}$ is linearly independent, lies in $W$, and has $m + 1$ elements, contradicting the maximality of $m$.

**Step 4.** So $S$ is a basis of $W$: $W$ is finite-dimensional and $\dim W = m \le n$.

**Step 5.** (b) If $\dim W = n$ then $S$ has $n$ elements and is linearly independent in $V$. Since $\dim V = n$, an independent set of $n$ elements is a basis of $V$, so $\operatorname{span} S = V$. But $\operatorname{span} S = W$. Hence $W = V$.

**Answer:** Maximal independent subsets of $W$ have size $m \le n$ and are bases of $W$, so $\dim W \le n$; if $m = n$ the basis of $W$ is an independent $n$-set in $V$, hence a basis of $V$, so $W = V$.

**19.** [Intermediate]

Let $W_1, W_2$ be subspaces of a vector space $V$. (a) Prove that $W_1 \cap W_2$ is a subspace of $V$. (b) Prove that $W_1 \cup W_2$ is a subspace of $V$ if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

**Step 1.** (a) *Zero:* $\mathbf{0} \in W_1$ and $\mathbf{0} \in W_2$ (both are subspaces), so $\mathbf{0} \in W_1 \cap W_2$.

**Step 2.** *Sums:* if $u, v \in W_1 \cap W_2$ then $u, v \in W_1$, so $u+v \in W_1$; likewise $u+v \in W_2$. So $u+v \in W_1 \cap W_2$. *Scalars:* for $c$ in the field, $cu \in W_1$ and $cu \in W_2$, so $cu \in W_1 \cap W_2$. Hence $W_1 \cap W_2$ is a subspace.

**Step 3.** (b) ($\Leftarrow$) If $W_1 \subseteq W_2$ then $W_1 \cup W_2 = W_2$, a subspace. Similarly if $W_2 \subseteq W_1$.

**Step 4.** ($\Rightarrow$) Suppose $W_1 \cup W_2$ is a subspace and $W_1 \not\subseteq W_2$. Then pick $u \in W_1$ with $u \notin W_2$. We show $W_2 \subseteq W_1$. Let $w \in W_2$ be arbitrary.

**Step 5.** Both $u$ and $w$ lie in $W_1 \cup W_2$, a subspace, so $u + w \in W_1 \cup W_2$. If $u + w \in W_2$, then $u = (u + w) - w \in W_2$ (as $w \in W_2$ and $W_2$ is a subspace), contradicting $u \notin W_2$. So $u + w \in W_1$.

**Step 6.** Then $w = (u + w) - u \in W_1$ (as $u \in W_1$). Since $w \in W_2$ was arbitrary, $W_2 \subseteq W_1$.

**Answer:** (a) closure of $W_1, W_2$ gives closure of $W_1 \cap W_2$ and $\mathbf{0} \in W_1 \cap W_2$. (b) $\Leftarrow$ trivial; $\Rightarrow$: if $u \in W_1 \setminus W_2$ and $w \in W_2$ then $u + w \in W_1$, hence $w \in W_1$; so $W_2 \subseteq W_1$.

**20.** [Intermediate]

Let $S_3$ be the set of symmetric matrices in $M_3(\mathbb{R})$ ($A^T = A$) and $K_3$ the set of skew-symmetric matrices ($A^T = -A$). Find a basis and the dimension of each, proving that your sets are bases.

**Step 1.** *Symmetric.* $A \in S_3$ means $a_{ij} = a_{ji}$ for all $i, j$. So $A$ is determined by $a_{11}, a_{22}, a_{33}, a_{12}, a_{13}, a_{23}$ and $A = a_{11}E_{11} + a_{22}E_{22} + a_{33}E_{33} + a_{12}(E_{12} + E_{21}) + a_{13}(E_{13} + E_{31}) + a_{23}(E_{23} + E_{32})$.

**Step 2.** So the six matrices $E_{11}, E_{22}, E_{33}, E_{12} + E_{21}, E_{13} + E_{31}, E_{23} + E_{32}$ (each symmetric) span $S_3$. They are independent: in a combination equal to 0, the entry in position $(1,1)$ is the coefficient of $E_{11}$, which must be 0; likewise for $(2,2), (3,3)$; position $(1,2)$ is the coefficient of $E_{12} + E_{21}$, and $(1,3)$, $(2,3)$ give the other two. All six coefficients are 0. So $\dim S_3 = 6$.

**Step 3.** *Skew-symmetric.* $a_{ij} = -a_{ji}$. For $i = j$: $a_{ii} = -a_{ii}$, so $2a_{ii} = 0$ and $a_{ii} = 0$ (we are over $\mathbb{R}$, where $2 \neq 0$). So $A$ is determined by $a_{12}, a_{13}, a_{23}$ and $A = a_{12}(E_{12} - E_{21}) + a_{13}(E_{13} - E_{31}) + a_{23}(E_{23} - E_{32})$.

**Step 4.** So $E_{12} - E_{21},\ E_{13} - E_{31},\ E_{23} - E_{32}$ (each skew-symmetric) span $K_3$. They are independent: in a combination equal to 0 the entries in positions $(1,2), (1,3), (2,3)$ are the three coefficients, so all are 0. So $\dim K_3 = 3$.

**Step 5.** Consistency check: $\dim S_3 + \dim K_3 = 6 + 3 = 9 = \dim M_3(\mathbb{R})$.

**Answer:** $S_3$: basis $E_{11}, E_{22}, E_{33}, E_{12}+E_{21}, E_{13}+E_{31}, E_{23}+E_{32}$, dim $= 6$. $K_3$: basis $E_{12}-E_{21}, E_{13}-E_{31}, E_{23} - E_{32}$, dim $= 3$.

**21.** [Challenge]

Let $W = \{A \in M_2(\mathbb{C}) :\ A^* = A\}$ (here $A^* = \overline{A}^T$ is the conjugate transpose). (a) Show that $W$ is a vector space over $\mathbb{R}$ and find a basis and $\dim_{\mathbb{R}} W$. (b) Show that $W$ is *not* a subspace of $M_2(\mathbb{C})$ over $\mathbb{C}$.

**Step 1.** (a) Write $A = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ with $p, q, r, s \in \mathbb{C}$. Then $A^* = \begin{pmatrix} \overline{p} & \overline{r} \\ \overline{q} & \overline{s} \end{pmatrix}$. So $A^* = A$ means $p = \overline{p}$, $s = \overline{s}$ and $q = \overline{r}$ (the $(2,1)$ condition $r = \overline{q}$ is the same). Hence $p, s \in \mathbb{R}$ and $r = \overline{q}$.

**Step 2.** Subspace over $\mathbb{R}$: $(A + B)^* = A^* + B^* = A + B$ and, for real $c$, $(cA)^* = \overline{c}A^* = cA$, and $0^* = 0$. So $W$ is closed under addition and real scalars and contains 0.

**Step 3.** Parametrize: $p = a$, $s = d$, $q = b + ci$ with $a, b, c, d \in \mathbb{R}$, so $r = \overline{q} = b - ci$. Then

$$A = \begin{pmatrix} a & b+ci \\ b-ci & d \end{pmatrix} = a\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + d\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + b\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + c\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}.$$

**Step 4.** So $H_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $H_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, $H_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $H_4 = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$ (all in $W$) span $W$ over $\mathbb{R}$.

**Step 5.** Independence over $\mathbb{R}$: if $aH_1 + dH_2 + bH_3 + cH_4 = 0$ with $a, b, c, d$ real, the diagonal gives $a = 0$, $d = 0$, and the $(1,2)$ entry gives $b + ci = 0$, so $b = c = 0$ (real and imaginary parts). So $\{H_1, H_2, H_3, H_4\}$ is a basis and $\dim_{\mathbb{R}} W = 4$.

**Step 6.** (b) $I_2 \in W$ since $I_2^* = I_2$. But $i\,I_2 = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}$ has $(iI_2)^* = \begin{pmatrix} -i & 0 \\ 0 & -i \end{pmatrix} = -iI_2 \neq iI_2$. So $iI_2 \notin W$: $W$ is not closed under multiplication by the complex scalar $i$, so it is not a complex subspace.

**Answer:** (a) $W$ is a real vector space with basis $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$, $\dim_{\mathbb{R}} W = 4$.
(b) $I_2 \in W$ but $(iI_2)^* = -iI_2 \neq iI_2$.

**22.** [Challenge]

For which real numbers $k$ is $\{(1,1,k),\ (1,k,1),\ (k,1,1)\}$ a basis of $\mathbb{R}^3$?

**Step 1.** Three vectors in $\mathbb{R}^3$ form a basis exactly when they are linearly independent (an independent set of $\dim V = 3$ elements is a basis), i.e. when the matrix with these columns has a pivot in every column. Row reduce, keeping $k$ as a symbol: $A = \begin{pmatrix} 1 & 1 & k \\ 1 & k & 1 \\ k & 1 & 1 \end{pmatrix}$.

**Step 2.** $R_2 \to R_2 - R_1$, $R_3 \to R_3 - kR_1$: $\begin{pmatrix} 1 & 1 & k \\ 0 & k-1 & 1-k \\ 0 & 1-k & 1-k^2 \end{pmatrix}$.

**Step 3.** $R_3 \to R_3 + R_2$: the $(3,3)$ entry becomes $(1-k^2)+(1-k) = (1-k)\big((1+k)+1\big) = (1-k)(k+2)$, and the $(3,2)$ entry becomes $(1-k)+(k-1) = 0$. So $\begin{pmatrix} 1 & 1 & k \\ 0 & k-1 & 1-k \\ 0 & 0 & (1-k)(k+2) \end{pmatrix}$.

**Step 4.** Case $k = 1$: the vectors are all $(1,1,1)$, so they are dependent (the matrix above has two zero rows). Not a basis.

**Step 5.** Case $k \neq 1$: the pivot $k-1$ in column 2 is nonzero. Column 3 has a pivot iff $(1-k)(k+2) \neq 0$, i.e. iff $k \neq 1$ and $k \neq -2$. So for $k \neq 1$ there are three pivots iff $k \neq -2$.

**Step 6.** Case $k = -2$: the third row is zero, so only two pivots. Directly, $(1,1,-2) + (1,-2,1) + (-2,1,1) = (0,0,0)$, a nontrivial relation, so they are dependent.

**Step 7.** Check one value, $k = 0$: the vectors $(1,1,0), (1,0,1), (0,1,1)$; the matrix above becomes $\begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ with three pivots, a basis ✓.

**Answer:** A basis exactly when $k \neq 1$ and $k \neq -2$ (the pivots are $1,\ k-1,\ (1-k)(k+2)$).

**23.** [Challenge]

Let $V$ be a vector space with $\dim V = n$ and let $S \subseteq V$ have exactly $n$ elements. Prove: (a) if $S$ is linearly independent then $S$ is a basis of $V$; (b) if $S$ spans $V$ then $S$ is a basis of $V$. You may use: the Extension Lemma, the Dimension Bound, and that every basis of $V$ has exactly $n$ elements.

**Step 1.** (a) Suppose $S$ is independent and, for contradiction, does not span $V$. Choose $x \in V$ with $x \notin \operatorname{span} S$.

**Step 2.** By the Extension Lemma, $S \cup \{x\}$ is linearly independent. It has $n+1$ elements (note $x \notin S$ because $S \subseteq \operatorname{span} S$).

**Step 3.** But $V$ is spanned by a basis, which has $n$ elements, so by the Dimension Bound every independent set in $V$ has at most $n$ elements. This contradicts $n+1$ elements. Hence $S$ spans $V$, and being independent, $S$ is a basis.

**Step 4.** (b) Suppose $S = \{v_1, \dots, v_n\}$ spans $V$ and, for contradiction, is linearly dependent: $c_1 v_1 + \cdots + c_n v_n = \mathbf{0}$ with some $c_j \neq 0$.

**Step 5.** Since $c_j \neq 0$ in a field we may divide: $v_j = -c_j^{-1} \sum_{i \neq j} c_i v_i$, so $v_j \in \operatorname{span}(S \setminus \{v_j\})$. Therefore $\operatorname{span}(S \setminus \{v_j\}) = \operatorname{span} S = V$ (any combination using $v_j$ can be rewritten without it).

**Step 6.** So $V$ is spanned by $n-1$ vectors. Then by the Dimension Bound every independent set in $V$ has at most $n-1$ elements; but a basis of $V$ is independent with $n$ elements. Contradiction. Hence $S$ is independent, and being spanning, a basis.

**Answer:** (a) A non-spanning independent $n$-set extends to an independent $(n+1)$-set, exceeding the bound $n$. (b) A dependent spanning $n$-set can drop a vector and still span, so $V$ would be spanned by $n-1$ vectors, forcing independent sets to have at most $n-1$ elements, contradicting a basis of size $n$.

**24.** [Challenge]

Let $S$ be a linearly independent subset of a vector space $V$ and let $S' \subsetneq S$ be a proper subset. Prove that $\operatorname{span}(S') \neq \operatorname{span}(S)$.

**Step 1.** Since $S' \neq S$ and $S' \subseteq S$, pick $v \in S$ with $v \notin S'$. Then $v \in \operatorname{span}(S)$.

**Step 2.** Claim: $v \notin \operatorname{span}(S')$. Suppose instead $v = c_1 s_1 + \cdots + c_m s_m$ with $s_1, \dots, s_m \in S'$ distinct (a combination of finitely many elements of $S'$; if $S' = \varnothing$ then $\operatorname{span} S' = \{\mathbf{0}\}$ and $v = \mathbf{0}$ is the empty combination).

**Step 3.** Then $1 \cdot v - c_1 s_1 - \cdots - c_m s_m = \mathbf{0}$ is a linear combination of distinct elements $v, s_1, \dots, s_m$ of $S$ (they are distinct because $v \notin S'$), and the coefficient of $v$ is $1 \neq 0$. This contradicts the linear independence of $S$. (If $S' = \varnothing$ the relation reads $v = \mathbf{0}$, impossible because an independent set contains no zero vector.)

**Step 4.** So $v \in \operatorname{span}(S)$ but $v \notin \operatorname{span}(S')$, hence the two spans differ.

**Answer:** An element $v \in S \setminus S'$ lies in $\operatorname{span} S$; if it were in $\operatorname{span} S'$ we would get a nontrivial relation among elements of $S$, contradicting independence.

**25.** [Challenge]

Let $V$ be a vector space over $\mathbb{C}$ and $u, v, w \in V$. (a) Prove $\{u, v, w\}$ is linearly independent if and only if $\{u+v,\ u+w,\ v+w\}$ is. (b) Where did you use $2 \neq 0$? (c) Show that the statement fails over the field $\mathbb{F}_2 = \{0, 1\}$ (where $1 + 1 = 0$): give $u, v, w \in \mathbb{F}_2^3$.

**Step 1.** Put $x = u + v$, $y = u + w$, $z = v + w$.

**Step 2.** (a) ($\Rightarrow$) Assume $\{u, v, w\}$ independent. Suppose $\alpha x + \beta y + \gamma z = \mathbf{0}$. Expand: $\alpha(u+v) + \beta(u+w) + \gamma(v+w) = (\alpha+\beta)u + (\alpha+\gamma)v + (\beta+\gamma)w = \mathbf{0}$.

**Step 3.** By independence of $u, v, w$: $\alpha + \beta = 0$, $\alpha + \gamma = 0$, $\beta + \gamma = 0$. Subtracting the first two gives

$\beta = \gamma$; then the third gives $2\beta = 0$, so $\beta = 0$ (here $2 \neq 0$ is used), hence $\gamma = 0$ and $\alpha = 0$. So $\{x, y, z\}$ is independent.

**Step 4.** ($\Leftarrow$) Assume $\{x, y, z\}$ independent. Solve for $u, v, w$: $x+y-z = (u+v)+(u+w)-(v+w) = 2u$, so $u = \frac{1}{2}(x + y - z)$; similarly $v = \frac{1}{2}(x - y + z)$ and $w = \frac{1}{2}(-x + y + z)$ (division by 2 is allowed since $2 \neq 0$).

**Step 5.** Suppose $au + bv + cw = \mathbf{0}$. Substitute: $\frac{1}{2}\big[(a + b - c)x + (a - b + c)y + (-a + b + c)z\big] = \mathbf{0}$. By independence of $x, y, z$: $a + b - c = 0$, $a - b + c = 0$, $-a + b + c = 0$. Adding the first two: $2a = 0$, so $a = 0$; adding the first and third: $2b = 0$, so $b = 0$; adding the last two: $2c = 0$, so $c = 0$. So $\{u, v, w\}$ is independent.

**Step 6.** (b) $2 \neq 0$ was used to conclude $2\beta = 0 \Rightarrow \beta = 0$ in ($\Rightarrow$), and to divide by 2 and to cancel $2a = 0$ in ($\Leftarrow$). In $\mathbb{C}$ (or $\mathbb{R}$) $2 \neq 0$ holds.

**Step 7.** (c) In $\mathbb{F}_2^3$ take $u = e_1 = (1, 0, 0)$, $v = e_2 = (0, 1, 0)$, $w = e_3 = (0, 0, 1)$: independent. Then $u + v = (1, 1, 0)$, $u + w = (1, 0, 1)$, $v + w = (0, 1, 1)$ and their sum is $(2, 2, 2) = (0, 0, 0)$ in $\mathbb{F}_2^3$. So $(u + v) + (u + w) + (v + w) = \mathbf{0}$ is a nontrivial relation: $\{u + v, u + w, v + w\}$ is dependent although $\{u, v, w\}$ is independent.

**Answer:** True over $\mathbb{C}$ (both directions need $2 \neq 0$). Over $\mathbb{F}_2$: $e_1, e_2, e_3$ are independent but $(e_1 + e_2) + (e_1 + e_3) + (e_2 + e_3) = \mathbf{0}$.

**26.** [Intermediate]

Let $S = \{v_1, \ldots, v_n\}$ be a set of $n$ distinct vectors in a vector space $V$. Prove that $S$ is a basis of $V$ if and only if every vector of $V$ can be written *uniquely* as $a_1v_1 + \cdots + a_nv_n$.

**Step 1.** ($\Rightarrow$) Let $S$ be a basis and $x \in V$. Since $S$ spans $V$, there are scalars with $x = \sum a_i v_i$ (existence).

**Step 2.** Uniqueness: if also $x = \sum b_i v_i$, subtract: $\sum(a_i - b_i)v_i = \mathbf{0}$. By linear independence every coefficient is 0: $a_i = b_i$ for all $i$.

**Step 3.** ($\Leftarrow$) Assume every $x \in V$ has a unique representation. Existence of a representation for every $x$ says $\operatorname{span} S = V$, so $S$ spans.

**Step 4.** Independence: $\mathbf{0} = 0 \cdot v_1 + \cdots + 0 \cdot v_n$ is one representation of $\mathbf{0}$. If $\sum c_i v_i = \mathbf{0}$ is another, uniqueness gives $c_i = 0$ for all $i$. So the only relation is the trivial one: $S$ is independent.

**Step 5.** So $S$ spans and is independent: it is a basis.

**Answer:** Spanning gives existence, independence gives uniqueness (subtract two representations); conversely existence is spanning, and uniqueness of the representation of $\mathbf{0}$ is independence.

**27.** [Challenge]

Let $P(\mathbb{R})$ be the vector space of all real polynomials (of any degree). Prove that $P(\mathbb{R})$ is not finite-dimensional.

**Step 1.** Suppose $P(\mathbb{R})$ has a finite basis with $m$ elements, so it is spanned by $m$ vectors.

**Step 2.** Consider $T = \{1, x, x^2, \ldots, x^m\} \subseteq P(\mathbb{R})$, a set of $m + 1$ polynomials.

**Step 3.** $T$ is linearly independent: suppose $c_0 + c_1x + \cdots + c_mx^m = 0$ as a polynomial. A nonzero polynomial of degree at most $m$ has at most $m$ roots, but the zero polynomial vanishes at every real number (infinitely many); so all $c_k = 0$.

**Step 4.** By the Dimension Bound, a linearly independent subset of a space spanned by $m$ vectors has at most $m$ elements. But $T$ is independent with $m + 1$ elements. Contradiction.

**Step 5.** Hence $P(\mathbb{R})$ is infinite-dimensional.

**Answer:** $\{1, \ldots, x^m\}$ is independent with $m + 1 > m$ elements: contradiction.